

Module 2 Assignment

Saturday, January 31, 2026

Registers Table

Stage	Process	Opcode (Bin)	PC	AC	IR	MAR	MBR
1A	FETCH @78A: MAR←PC, MBR←M[MAR], IR←MBR, PC←PC+1	—	78B	0000	78A0	78A	78A0
1B	EXECUTE: 0111 LOAD from 8A0: MAR←8A0, MBR←M[8A0], AC←MBR	0111	78B	111F	78A0	8A0	111F
2A	FETCH @78B	—	78C	111F	88A1	78B	88A1
2B	EXECUTE: 1000 ADD from 8A1: MAR←8A1, MBR←M[8A1], AC←AC+MBR	1000	78C	1B3A	88A1	8A1	0A1B
3A	FETCH @78C	—	78D	1B3A	88A2	78C	88A2
3B	EXECUTE: 1000 ADD from 8A2: MAR←8A2, MBR←M[8A2], AC←AC+MBR	1000	78D	3677	88A2	8A2	1B3D
4A	FETCH @78D	—	78E	3677	28A2	78D	28A2
4B	EXECUTE: 0010 STORE to 8A2: MAR←8A2, MBR←AC, M[8A2]←MBR	0010	78E	3677	28A2	8A2	3677

Stage 1A (after FETCH 1)

Address	Contents
78A	78A0
78B	88A1
78C	88A2
78D	28A2
8A0	111F
8A1	0A1B
8A2	1B3D

MAR ← PC
MBR ← M[MAR]
IR ← MBR
PC ← PC + 1

Stage 1B (after EXECUTE 1)

Address	Contents
78A	78A0
78B	88A1
78C	88A2
78D	28A2
8A0	111F
8A1	0A1B
8A2	1B3D

MAR ← IR [Address] = 8A0
MBR ← M[MAR] = M[8A0]
AC ← MBR

Stage 2A (after FETCH 2)

Address	Contents
78A	78A0
78B	88A1
78C	88A2
78D	28A2
8A0	111F
8A1	0A1B
8A2	1B3D

MAR ← PC
MBR ← M[MAR]
IR ← MBR
PC ← PC + 1

Stage 2B (after EXECUTE 2)

Address	Contents
78A	78A0
78B	88A1
78C	88A2
78D	28A2
8A0	111F
8A1	0A1B
8A2	1B3D

MAR ← IR [Addr] = 8A1
MBR ← M[MAR] ⇒
⇒ M[8A1] = 0A1B
AC ← AC + MBR
AC = 111F
MBR = 0A1B
111F + 0A1B = 1B3A

Stage 3A (after FETCH 3)

Address	Contents
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Stage 3B (after EXECUTE 3)

Address	Contents
78A	78A0
78B	88A1
78C	88A2
78D	28A2
8A0	111F
8A1	0A1B
8A2	1B3D

MAR ← 8A2

Stage 3A (after FETCH 3)

Address Contents

78A	78A0
78B	88A1
78C	88A2
78D	28A2
8A0	111F
8A1	0A1B
8A2	1B3D

Stage 3B (after EXECUTE 3)

Address Contents

78A	78A0
78B	88A1
78C	88A2
78D	28A2
8A0	111F
8A1	0A1B
8A2	1B3D

$MAR \leftarrow 8A2$
 $MBR \leftarrow M[8A2] = 1B3D$
 $AC = AC + MBR$

$AC = 1B3A$
 $MBR = 1B3D$
 $1B3A + 1B3D = 3677$

Stage 4A (after FETCH 4)

Address Contents

78A	78A0
78B	88A1
78C	88A2
78D	28A2
8A0	111F
8A1	0A1B
8A2	1B3D

Stage 4B (after EXECUTE 4)

Address Contents

78A	78A0
78B	88A1
78C	88A2
78D	28A2
8A0	111F
8A1	0A1B
8A2	3677

$MAR \leftarrow 8A2$
 $MBR \leftarrow AC$
 $M[MBR] \leftarrow MBR$

$MBR \leftarrow 3677$
 $M[8A2] \leftarrow 3677$